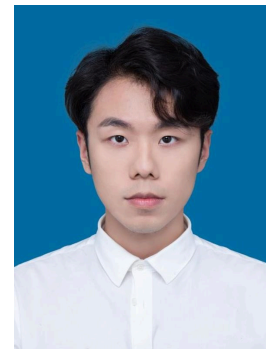


Chenyang Ai

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📍 Nanchang, Jiangxi Province, China 📅 1st July 2000 ♂ male



Research Interests

- NN Hardware/Software Co-design
- The tensor compiler and MLIR
- The microarchitecture of VPU, GPU, NPU, CGRA
- Design space exploring for chiplet

Education

September 2022 –
present
Beijing, China

Master Degree, Peking University

- major is Integrated Circuit Science and Engineering,
- Graduate Recommended Admission Qualification

September 2018 –
June 2022
Chongqing, China

Bachelor's Degree,

Chongqing University of Posts and Telecommunications

- major is Microelectronics Science and Engineering
- 3.79/4.0 rank 1/35

Publications

MPTVPU: A RISC-V VPU Could Solve the Problems of Precision and Tensor,

Chenyang Ai, Lechuan Zhao, Zhijie Huang, Cangyuan Li, Xinan Wang, Ying Wang

- 2025 [Design Automation Conference](#) ☑ (DAC)
(under review)
- <https://arxiv.org/abs/2405.02196>

What Is Relationships between Utilization and Generality of Systolic Array: Insights and Solution,

Chenyang Ai*, Lechuan Zhao*, Xinan Wang, Ying Wang

- 2025 Design Automation Conference (DAC) under review
-

HVSA: A Deeply Hybrid Vector Systolic Architecture with Dynamically Reconfigurable Dataflow,

Chenyang Ai*, Lechuan Zhao*, Xinan Wang, Ying Wang

- 2024 China Computer Federation Chip Conference (CCF CHIP'24 in Chinese) (This conference will not be published in the proceedings, but it is the largest chip design-related conference in China.)
- Journal of Computer Science and Technology (CCF-A in Chinese) Invited by CHIP'24 (35%) , further submission to this journal.

Projects

January 2020 –
August 2021

End-to-End Hardware Accelerators for LSTM,CNN,ATTENTION,
project leader

During my undergraduate, I created several datasets and trained for networks like CNN, LSTM, and attention network to recognize static and dynamic hand gesture and speech words. I have also designed specialized hardware to accelerate neural networks and audio/image processing by FPGA and applied them in real scenarios successfully. Therefore, I got top rankings at top competitions in China.

January 2023 – July 2024

Combine the Systolic Array and Vector Unit, project leader
two DAC 2025 papers

January 2024 – June 2024

Heterogeneous SoC Compiler Based on MLIR,
main participant/ranking second

Based on tensor contraction, I converted various tensor operators into matrix multiplication, and I have built an MLIR-based compilation chain. Therefore, the compiler could schedule the kernel to different accelerators to improve multi-thread performance and utilization.

March 2024 – present

Tensor Train (TT) Full-Process Accelerator for LLMs, project leader

To solve the problem of excessive weights in current large language models (LLMs), I plan to explore the Tensor Train (TT) decomposition methods and leverage previous systolic arrays works to develop a specialized accelerator for TT. Compared to existing hardware implementations, this architecture not only accelerates the TT decomposition but also supports all vector-matrix multiplication operations during the LLM's decoding.

April 2024 – present

The Analysis Framework for the Generality of ML Accelerators,
project leader

The current trend involves the adoption of various spatial architecture to accelerate specified computing kernels, such as systolic array for GEMM, eyeriss for CONV. The goal of this project is to develop a comprehensive analytical framework based on tensor algebra. This framework will allow us achieve the optimal hardware configuration within arbitrary range of tensor operators.

Awards

September 2021

**Third Ranking in Digital Track, The 3rd 'FudanMicroCup' 2021
National Student Electronic Design Competition,** Shanghai Fudan
Microelectronics Group Co., Ltd. School of Microelectronics, Fudan
University

Speech Recognition Accelerator Based on LSTM

July 2021

**First Ranking in Diligent Track, CHINA COLLEGE Integrated Circuit
COMPETITION,** Ministry of Industry and Information Technology Talent
Exchange Center, Chinese largest integrated circuit design competition
SoC for Sign Language and Speech Collaborative Recognition Based on CNN

References

Xin'an Wang, *Professor*, School of Integrated Circuits, Peking University
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Ying Wang, *Researcher*, Institute of Computing Technology, Chinese Academy of Sciences
wangying2009@ict.ac.cn

Hongsheng Zhang, *Associate Dean*,
School of Optoelectronic Engineering, Chongqing University of Posts and Telecommunications
zhanghs@cqupt.edu.cn